Robotic Manipulation

Exercise 5 - Report

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• The approach chosen for planning the grasp

Two contacts with friction can be modelled with 4 contacts without friction, with the help of friction cones. We assume that the friction coefficient is 1 for all objects: That means that the angle of the friction cones is $\tan^{-1}(1) = 45^{\circ}$. If the two grasping points are in each other's friction cones, then we can talk about stable grasping. In order to get the optimal grasping point, we have to look at all the stable grasps and choose the grasping point pair, which has the shortest distance from the origin of the polygon.

My Python script takes the data points and forms the polygon from these coordinates. Then I generate the points between each vertex and iterate over these points. For each possible grasping point, I calculate the friction cones with 45 degrees and form a triangle. I check, which points on the periphery of the polygon are inside this triangle and save this information into a dictionary. Once I iterated through every point, I check which grasping points have each other in the friction cone 'triangle' and create grasping point pairs. Each of them are considered as a **stable grasp**.

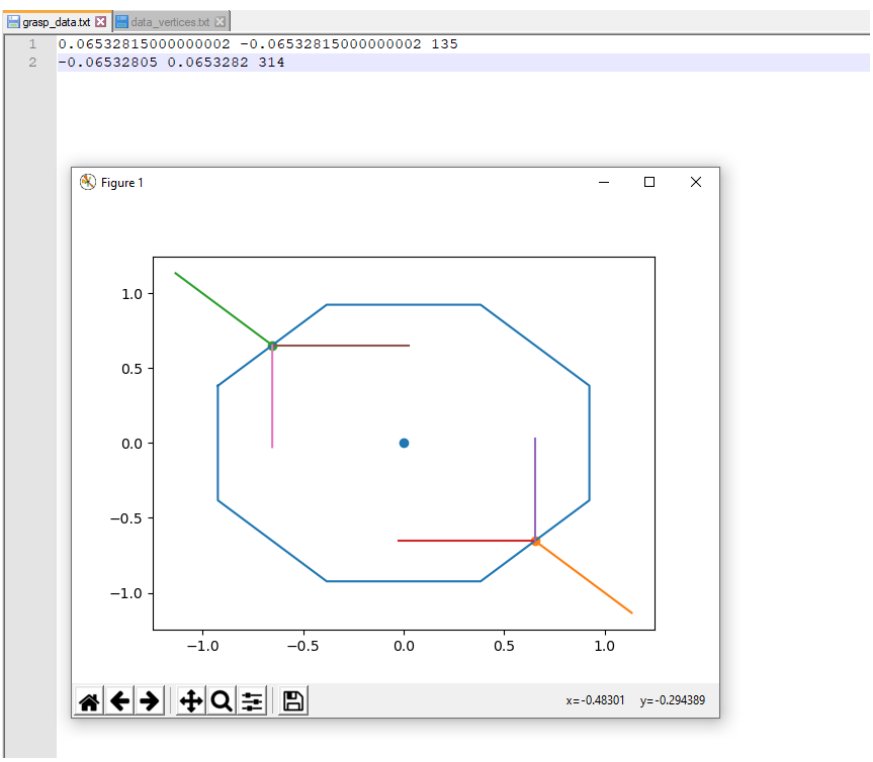
After I calculated the list of stable grasps, I iterate through these grasping point pairs and calculate the distance between the point and the center of the polygon to find the minimum distance. The grasping point pair that has the minimum distance from the center is the **optimal grasping point pair**. In case the polygon is symmetric, there are multiple optimal grasping points in 2D. If that is the case, I calculate the first optimal grasping point pair.

• How to run your grasp planning code

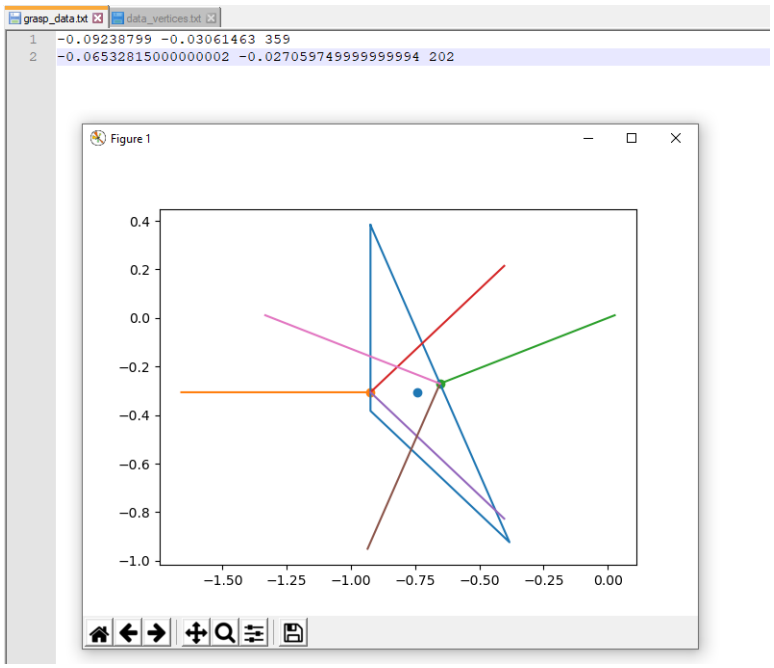
```
python3.7 roboticmanip5.py –input PATH\data_vertices.txt --output PATH\grasp_data.txt
```

I executed the script in the following way: Run the roboticmanip5.py script with 2 arguments: The first argument is the input, where we want the data points to be read from. The second argument is the output, where we want the grasping points and their angles to be written to.
When the code finished executing, the data points and their angles with respect to the x axis will be written into the specified output file. Furthermore, the calculated points with the approach vector and friction cones will be visible alongside with the polygon.

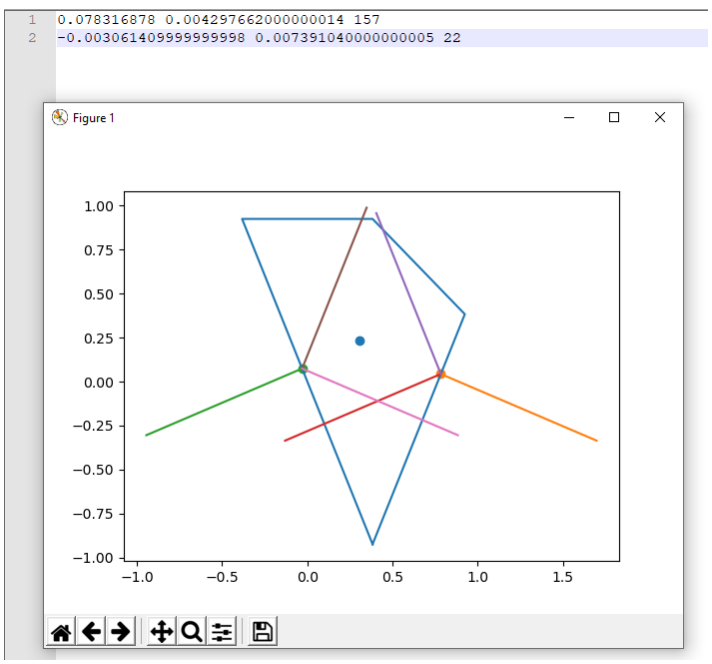
• Visualization of the grasping points, approach direction, and friction cones on the object polygon



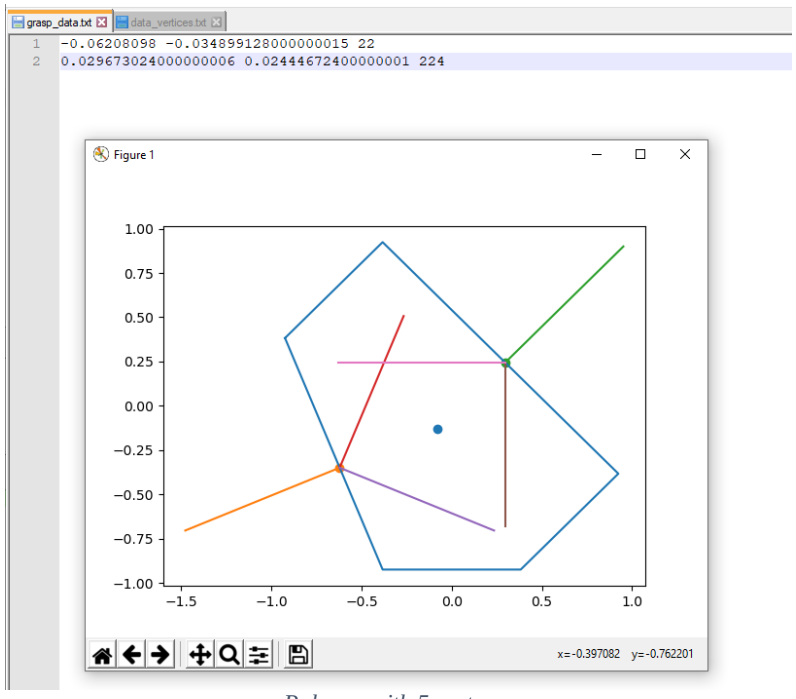
Original polygon



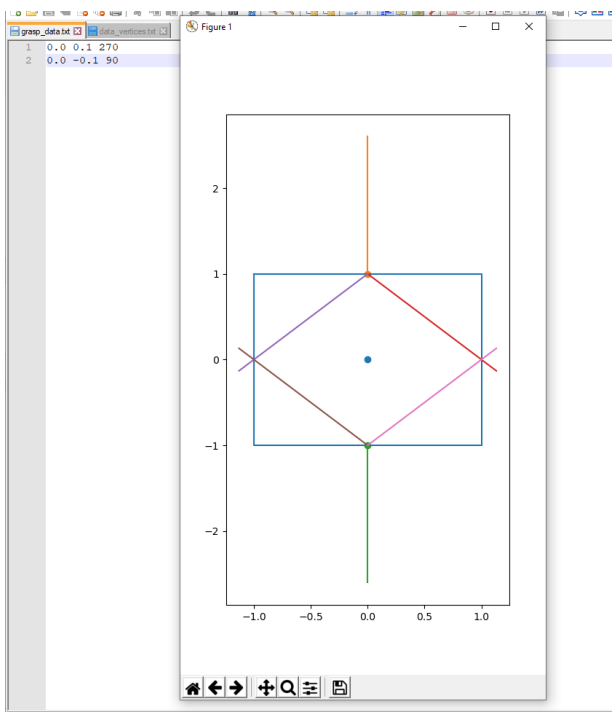
Polygon with 3 vertexes



Polygon with 4 vertexes



Polygon with 5 vertexes



Square

• Discussion of the results and answers to the following questions:

– Is the grasp both force-closure and form-closure, either of, or neither? Explain why.

If the fingertips are able to generate forces at the contacts so as to resist any external force or moment applied to the object, then the object is said to be in force closure. If the contacts are placed so as to make any motion of the object impossible, the object is then said to be in form closure. All form closure grasps are also force closure, but the reverse is not always true.
The two fingers with frictional point contacts are grasping a polygon in the presence of gravity. **The polygon is in force closure** because for any external wrench there exist contact wrenches that cancel it. However, if for example, the motors powering the fingers cannot provide sufficient forces, the polygon may fall despite the grasp being force closure, because it is not in form closure.

– If the object was not force-closure, where would we need to place additional contacts to achieve it?

According to the *theorem*: For any bounded shape that is not a surface of revolution, a force closure grasp exists (Mishra et al., 1987). That means, that with enough contact points we can achieve a stable grasp. [1]
According to another *theorem [1]:* A grasp with at least two distinct soft finger contacts is force closure if it is in **equilibrium**, with contact forces pointing strictly into the friction cones at the respective points of contact**. Assuming we have a third contact point, we should place it so that the grasp quality metrics are maximized and that its contact force is pointing into the friction cones at the respective points of contact**. We scale moments /torques/ by the characteristic length of the polygon from the origin of the polygon and find the shortest distance, because in that direction is the easiest to "steal" the object. [2]
Another approach could be the **sampling based grasp planning**. We choose candidate contacts and evaluate the resulting grasps by simulation. [2]

• For the bonus task specifically report the approach and the metric used for planning the stable grasp

For the optimal grasping point calculation, I used the approach we were taught in class. In order to find the optimal grasp amongst the stable grasps, I calculated the distances between the grasping points and the origin of the polygon. The grasping point pair with the minimum total distance is the **optimal grasping point pair**.

• Estimate of time spent on this exercise

24 hours.

References

- 1 - Force-closure Grasps With Two Palms
- 2- Robotic Manipulation lecture 7